Fault · target	Correct action	Runbook	PER-SAC (warm)	PER-SAC (cold)
F1 · catalog	RESTORE catalog	RESTORE catalog (10/10)	RESTORE catalog (10/10)	RESTORE cart (10/10) SCALE_UP frontend (2/10) ≈22 actions per episode
F2 · cart	RESTART cart	RESTART cart (5/5)	RESTART cart (5/5)	RESTORE cart (5/5) ≈22 actions per episode
F2 · currency	RESTART currency	RESTART currency (3/5) no action (2/5)	no action (5/5)	RESTORE cart (5/5) ≈23 actions per episode
F3 · cart	RESTORE cart	RESTORE cart (5/5) RESTART cart (1/5)	RESTORE cart (5/5)	RESTORE cart (5/5) ≈10 actions per episode
F3 · currency	RESTORE currency	RESTORE currency (5/5)	RESTORE currency (5/5)	RESTORE cart (5/5) ≈22 actions per episode
F4 · frontend	SCALE_UP frontend	SCALE_UP frontend (10/10)	SCALE_UP frontend (10/10)	SCALE_UP frontend (10/10) RESTORE cart (10/10) ≈10 actions per episode
NULL (no fault)	no action	no action (10/10)	no action (10/10)	RESTORE cart (10/10) ≈11 actions per episode